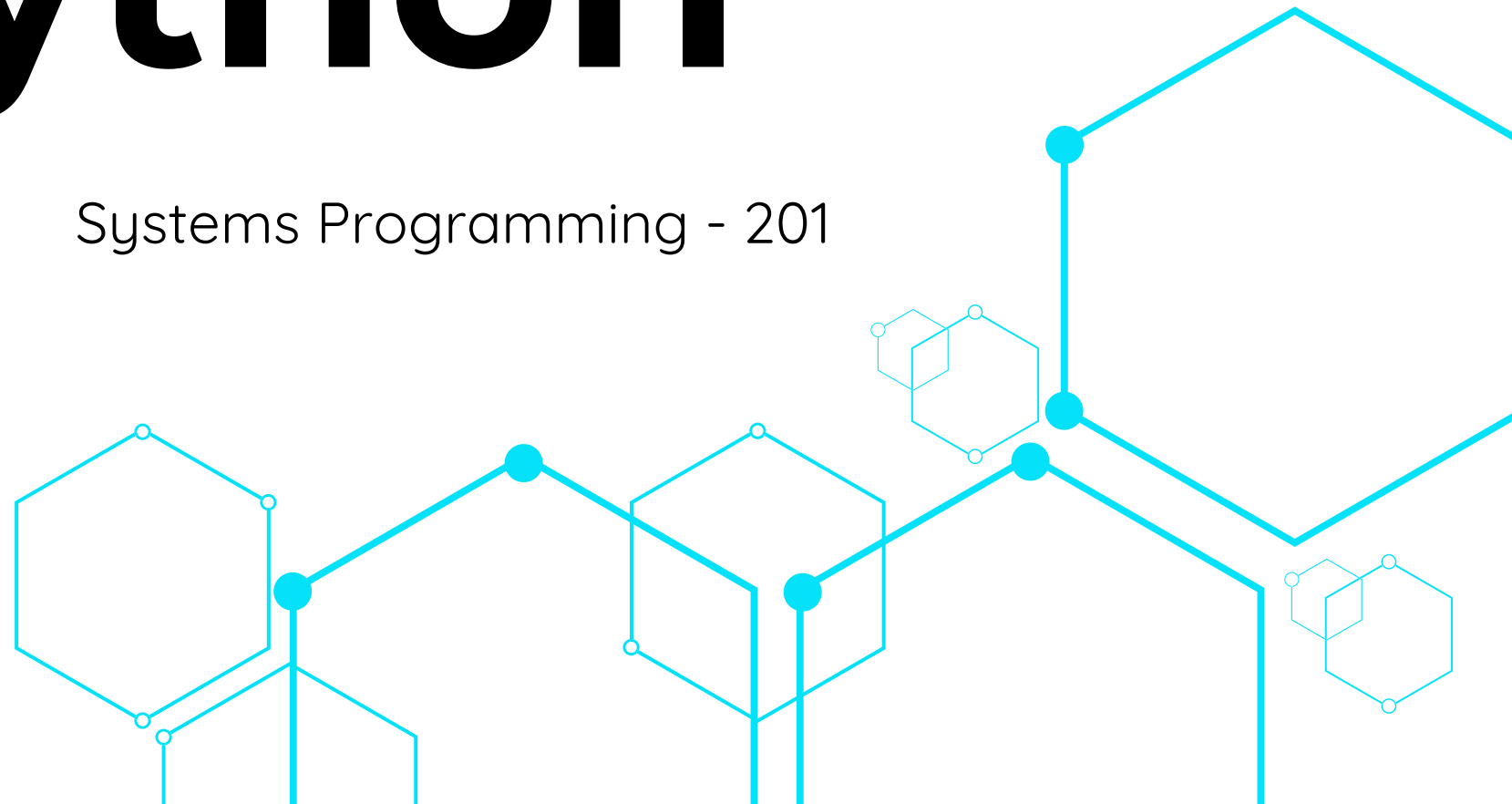


Mini Shell in Python

Systems Programming - 201



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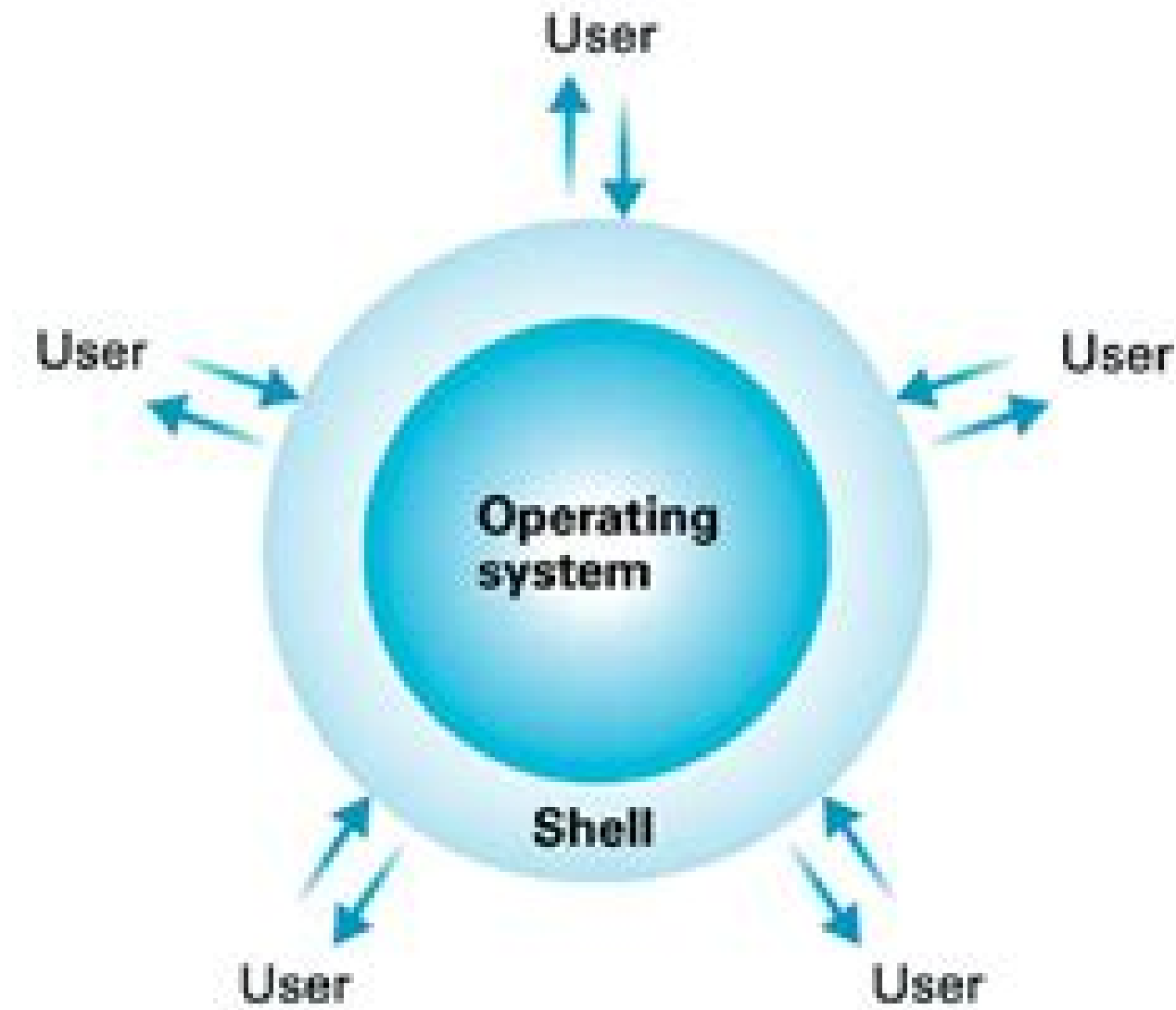
What is a Shell?

Content:

- A shell is a command-line interface (CLI)
- Allows users to communicate with the operating system
- Users type commands, computer executes them

Examples of shells:

- Bash (Linux/Mac)
- CMD (Windows)
- PowerShell (Windows)
- Our Mini Shell (Python)





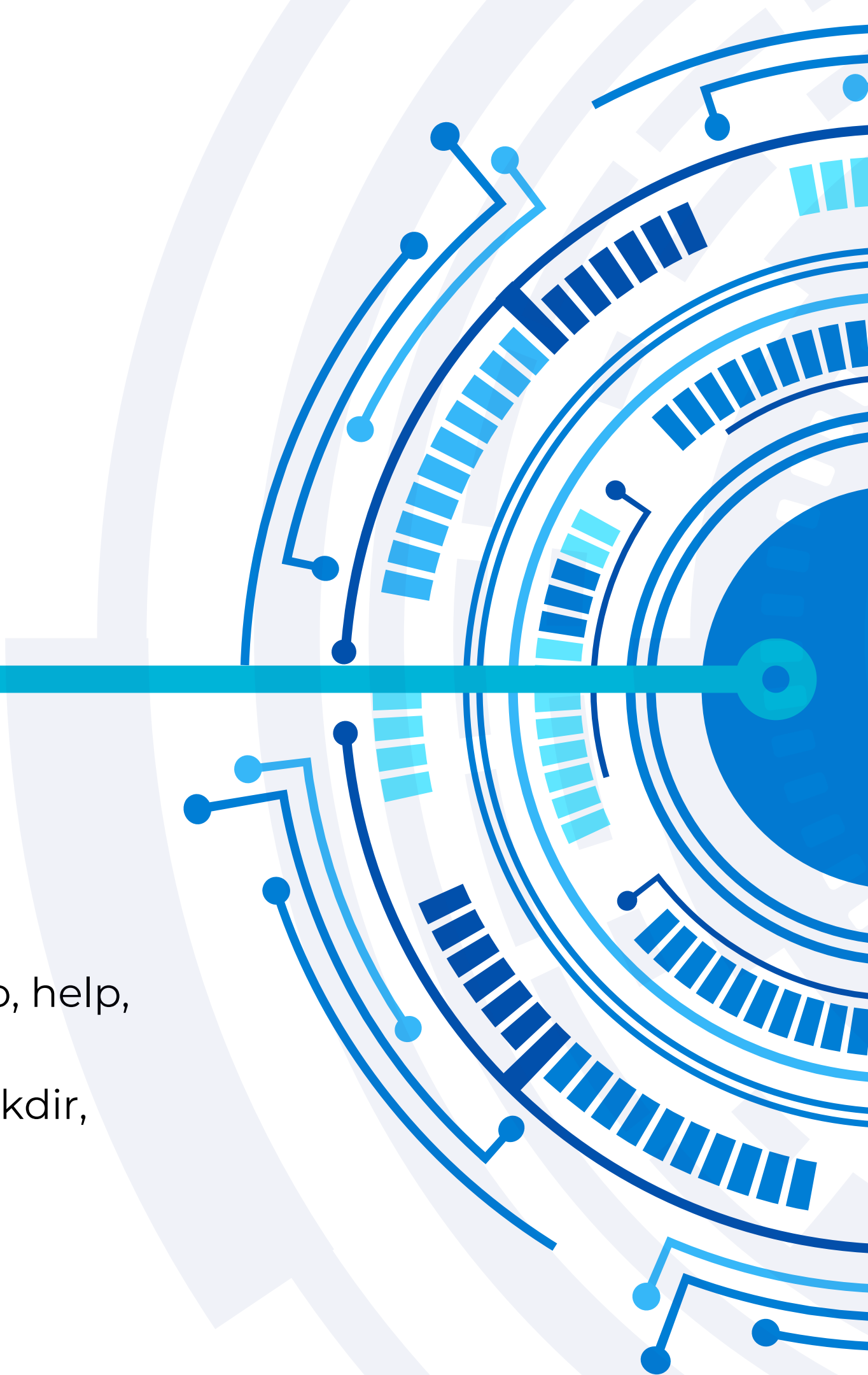
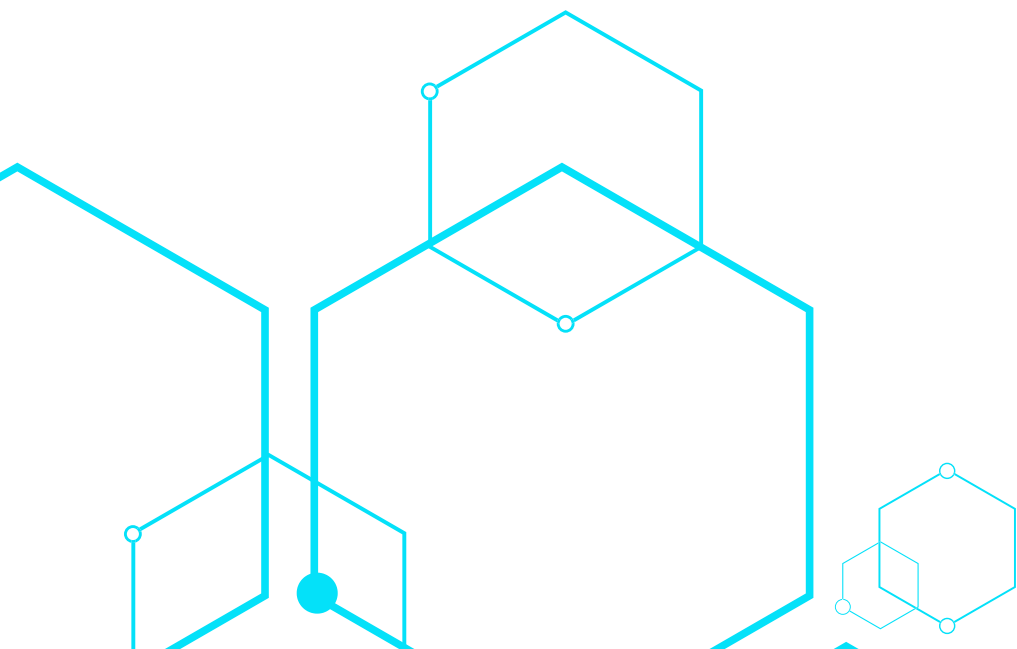
Motivation Behind the Project

Why Did We Build This?

- To understand how command-line shells work internally
- To learn systems programming concepts
- To improve Python programming skills
- To simulate a real operating system shell
- To apply course concepts (loops, conditionals, functions, imports)

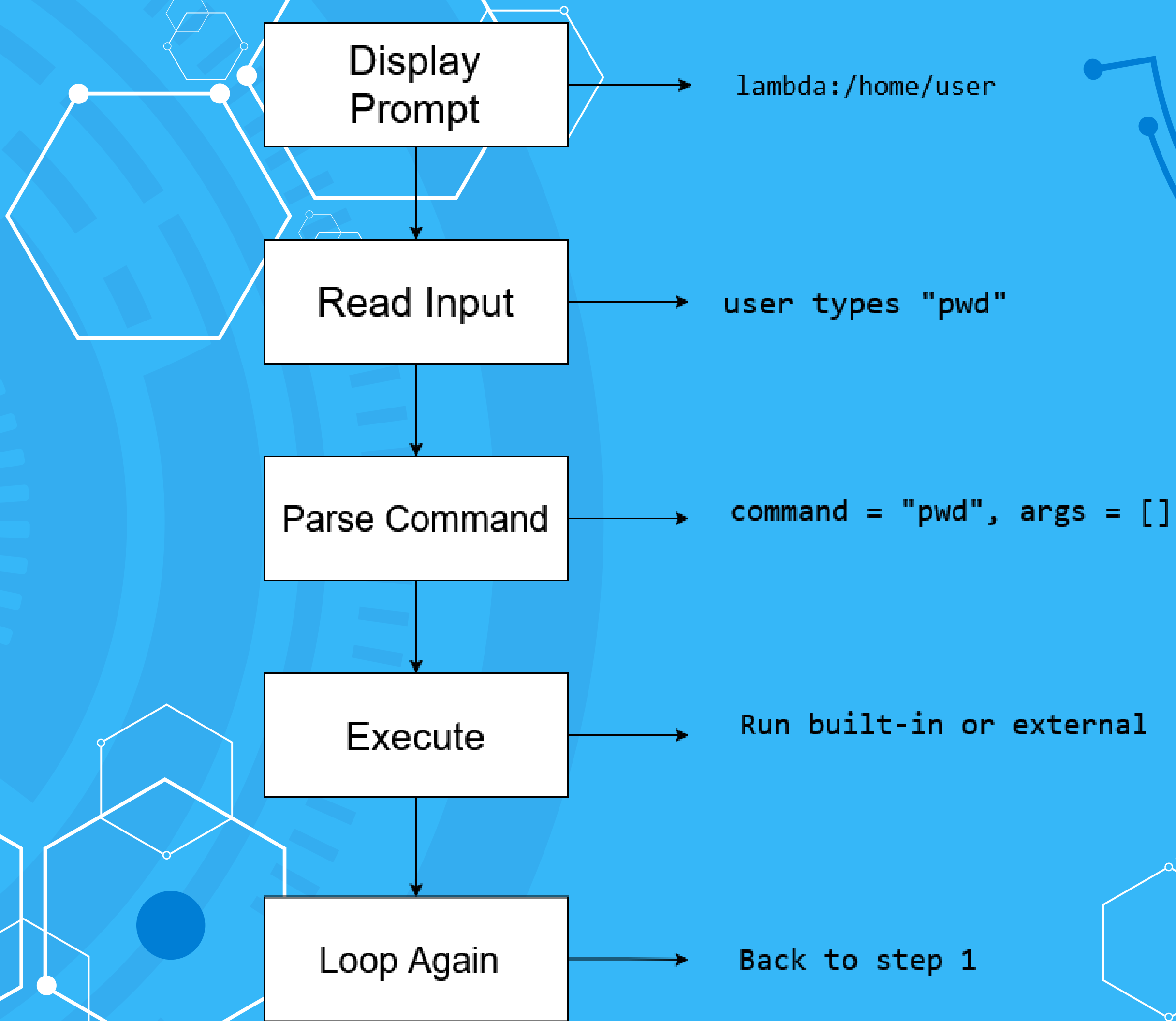


Project Objectives



What Our Shell Can Do?

- Read user commands from keyboard
- Parse commands and arguments
- Execute built-in commands (cd, pwd, echo, help, exit)
- Execute external system commands (ls, mkdir, cat, etc.)
- Handle errors gracefully without crashing
- Display current directory in the prompt



Built-in Commands

Example

```
lambda:/home/student$ pwd  
/home/student
```

```
lambda:/home/student$ cd Documents
```

```
lambda:/home/student/Documents$ pwd  
/home/student/Documents
```

```
lambda:/home/student/Documents$ cd
```

```
lambda:/home/student$
```

Command	Description
pwd	Shows your current folder path
cd folder	Moves into a folder
cd (by itself)	Goes to your home folder

Competitive Advantage

Example

```
lambda:/home$ echo Welcome to Lambda Shell!  
Welcome to Lambda Shell!  
  
lambda:/home$ help  
  
--- Available Commands ---  
cd [dir] - Change directory  
pwd      - Show current directory  
echo text - Print text  
help     - Show this help  
exit/quit - Exit the shell  
-----  
  
lambda:/home$ exit  
Goodbye!
```

Command	Description
echo [text]	Prints whatever you type
help	Shows all available commands
exit	Exit shell
quit	Exit shell

External Command

Running System Commands

Command	Description
ls	Lists all files in current folder
mkdir newfolder	Creates a new folder called "newfolder"
cat file.txt	Shows what's inside file.txt
rm file.txt	Deletes file.txt
fakecommand	Shows: Error: 'fakecommand' is not recognized

How Our Shell Handles Errors



What user does	Our shell's response
Types nothing, just presses Enter	Shell ignores it and waits for real command
Types cd fakefolder	Error: Directory 'fakefolder' not found
Types wrongcommand	Error: 'wrongcommand' is not recognized
Types exit or quit	Goodbye! and closes

```
else:
    try:
        subprocess.call(user_input, shell=True)
    except:
        print(f"Error: '{command}' is not recognized")
```

```
os.chdir(os.path.expanduser("~"))
else:
    try:
        os.chdir(args[0])
    except FileNotFoundError:
        print(f"Error: Directory '{args[0]}' not found")
```

Code Demo

```
kali@kali: ~  
Session Actions Edit View Help  
=====
```

Welcome to Lambda Mini Shell!
Type 'exit' or 'quit' to leave

```
=====
```

lambda:/home/kali\$ pwd
/home/kali
lambda:/home/kali\$ cd Documents
lambda:/home/kali/Documents\$ echo Hello! I'm Salma!
Hello! I'm Salma!
lambda:/home/kali/Documents\$ echo This is my Group's Demo
This is my Group's Demo
lambda:/home/kali/Documents\$ help

— Available Commands —
cd [dir] - Change directory
pwd - Show current directory
echo text - Print text
help - Show this help
exit/quit - Exit the shell

```
=====
```

lambda:/home/kali/Documents\$ quit
Goodbye!

Full Code

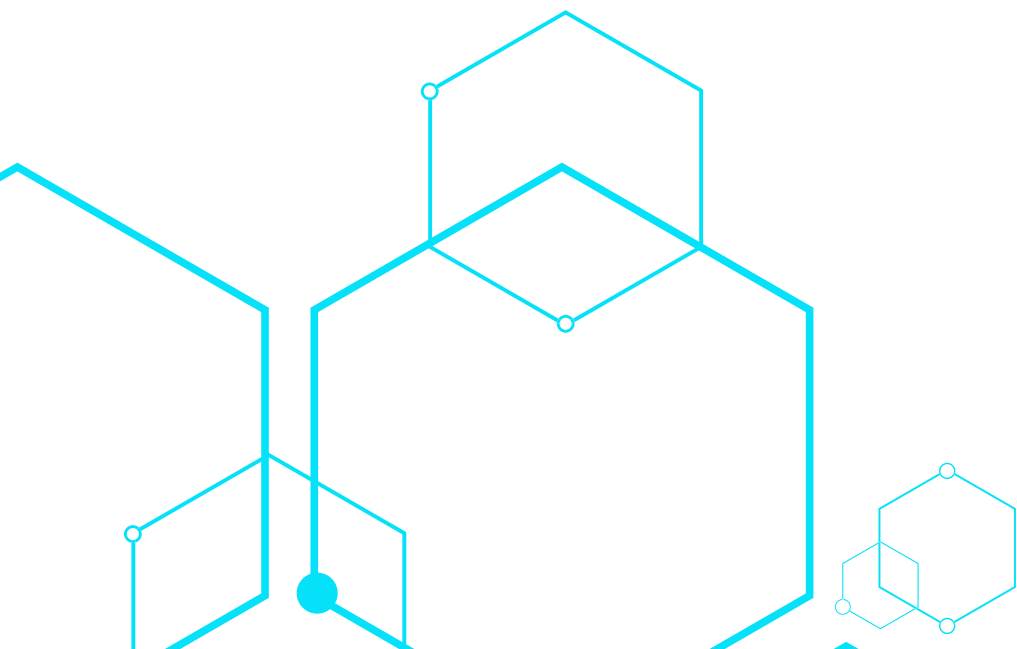
```
~/mini_shell.py - Mousepad
File Edit Search View Document Help

1 import os
2 import subprocess
3
4 print("\n" + "="*50)
5 print("  Welcome to Lambda Mini Shell!")
6 print("  Type 'exit' or 'quit' to leave")
7 print("="*50 + "\n")
8
9 while True:
10     current_dir = os.getcwd()
11
12     user_input = input(f"lambda:{current_dir}$ ")
13
14     user_input = user_input.strip()
15
16     if user_input == "":
17         continue
18
19     parts = user_input.split()
20     command = parts[0]
21     args = parts[1:]
22
23     if command == "exit" or command == "quit":
24         print("Goodbye!")
25         break
26
27     elif command == "cd":
28         if len(args) == 0:
29             os.chdir(os.path.expanduser("~/"))
30         else:
31             try:
32                 os.chdir(args[0])
33             except FileNotFoundError:
34                 print(f"Error: Directory '{args[0]}' not found")
35
36
37
```

```
38     elif command == "pwd":
39         print(os.getcwd())
40
41     elif command == "echo":
42         print(" ".join(args))
43
44     elif command == "help":
45         print("\n— Available Commands —")
46         print("cd [dir] - Change directory")
47         print("pwd      - Show current directory")
48         print("echo text - Print text")
49         print("help      - Show this help")
50         print("exit/quit - Exit the shell")
51         print("—————\n")
52
53     else:
54         try:
55             subprocess.call(user_input, shell=True)
56         except:
57             print(f"Error: '{command}' is not recognized")
58
59
60
```



Conclusion



- 
- Successfully implemented a mini shell in Python
 - Supports built-in and external commands
 - Demonstrates systems programming concepts

Future Improvements

- Add command history
- Add piping
- Add file redirection



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Thank You For Listening!

Any Questions?

Ask Away !